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strongcoca: A Python package for strong coupling in nanoscale systems

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Software

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Summary

Electromagnetic (EM) simulations are ubiquitous for the modeling of spectroscopy and near–field enhancement in plasmonics, connecting microscopic phenomena to experimentally observed quantities. Here, we present `strongcoca`, a Python package that calculates the EM response using a subsystem approach. In this approach, the response of a system is calculated from the interactions between subsystems and the individual response of each subsystem, e.g., a molecule or a nanostructure. The user supplies tabulated response functions for the subsystems as well as their positions and orientations. In the current implementation of `strongcoca`, the EM interactions are approximated up to the order of dipole–dipole interactions, as discussed further below.

Response functions for subsystems can be taken from various sources. For example, `strongcoca` provides parsers for the output of time-dependent density-functional theory (TDDFT) calculations carried out using the `GPAW` (Mortensen et al., 2024) and `NWChem` (Aprà et al., 2020) codes. It is also possible to use analytical Mie–Gans models with tabulated dielectric functions, which allows one to use experimental data. Additionally, `strongcoca` interfaces to the Atomic Simulation Environment (ASE) (Larsen et al., 2017) for visualization of the coupled systems and relies on `NumPy` (Harris et al., 2020) and `SciPy` (Virtanen et al., 2020) for linear algebra and vectorized operations.

The full documentation for `strongcoca` in addition to examples and tutorials can be found at https://strongcoca.materialsmodeling.org/.

Statement of need

In the common approach to EM simulations, matter is treated as a continuum, with material-specific response functions (e.g., dielectric functions) obtained from experiment or simulations of bulk materials. This approach fails to accurately capture the response of nanoscale materials with irregularities on the atomic scale, or of individual organic molecules, due to the inherent quantum nature of electrons on the atomic scale. On the other hand, fully quantum mechanical calculations, while accurate, exhibit steep polynomial scaling with system size and remain intractable even for moderately sized systems. In the approach to EM simulations taken in `strongcoca`, we instead rely on explicitly separating the system of interest into subsystems for which the response is known, thereby circumventing these issues. The response function (polarizability) of each subsystem can capture the correct quantum behavior if it is obtained at the appropriate level of theory or taken from experiment. strongcoca is therefore aimed at researchers in nanophotonics and quantum optics who need to model interactions between distinct quantum and nanoscale subsystems.

In the current implementation, strongcoca neglects (1) higher–order multipolar interactions between subsystems, (2) charge transfer between subsystems, and (3) quantum mechanical hybridization between the subsystems. These approximations are less significant for well-separated subsystems, but lead to inaccurate results at small separations. It is possible to extend strongcoca to properly take into account higher–order multipoles (1) and charge transfer (2), but the lack of quantum mechanical hybridization (3) is fundamentally a limitation of our subsystem approach. The dipole–dipole interactions currently implemented in strongcoca are sufficient to study, e.g., strong light–matter coupling (Ebbesen, 2016) between nanoscale materials and organic molecules. strongcoca is thus an excellent tool for **strong coupling calculations**, from which the package takes its name.

State of the field

Two EM codes whose implementation most closely resembles that of strongcoca are PyGDM2 (Wiecha, 2018; Wiecha et al., 2022), implementing the Green dyadic method, and DDSCAT (Draine & Flatau, 1994), implementing the discrete–dipole approximation. Both of these codes calculate the response of a system made up of many interacting dipoles that together represent a single continuum material. In strongcoca, by contrast, each dipole represents an explicit subsystem, such as an organic molecule or a nanostructure, with its own individually known response function. strongcoca therefore has an inherently different scope than these packages: rather than modeling the EM response of a single material, it is designed to study interactions between distinct subsystems with individually known response functions. However, in principle a single larger polarizable nanoparticle could also be described in strongcoca as a collection of closely packed dipolar elements, just as in other discrete–dipole approximation implementations, thereby enabling the description of higher-order multipolar interactions between this subsystem and others.

The code PyCharge (Filipovich & Hughes, 2022) can also calculate dipole–dipole interactions, although it supports only single-Lorentzian response functions, i.e., subsystems with a single resonance. While it would be possible to describe a general material using a sum of Lorentzians, there is no user interface to conveniently do so. In addition, PyCharge performs simulations in the time domain (allowing for explicitly time-dependent positions), while strongcoca works in the frequency domain, which is orders of magnitude faster for the applications targeted here.

Software design

strongcoca follows an object-oriented structure, where physical subsystems are defined independently and then assembled into an interacting system. The response of each subsystem is encapsulated in a PolarizableUnit object, which carries a response function, a position, and an orientation in space. The response function can be provided through any of the available parsers, for example from a TDDFT calculation in GPAW or NWChem, or from an analytical Mie–Gans model with a tabulated dielectric function. This makes it straightforward to mix subsystems described at different levels of theory or obtained from different software packages. Once the individual subsystems are defined, they can be combined into a CoupledSystem object, which manages the dipole–dipole interaction tensor between all pairs of subsystems. Physical observables of the interacting system, such as absorption spectra and near-field enhancements, are then computed by passing the CoupledSystem to the appropriate calculator object.

The object-oriented design also makes it straightforward to extend strongcoca with new parsers, new response function types, and new calculators, each of which can be added independently without modifying the core framework.

Research impact statement

strongcoca has been under active development for approximately five years, during which it has been used internally by the developers to study strong light–matter coupling in plasmonic systems. In (Fojt et al., 2021) it was used to study strong coupling in assemblies of molecules and nanoparticles, and in (Kuisma et al., 2022) to study strong coupling between molecules and plasmonic nanocavities. Following requests from other researchers, we have decided to release strongcoca as an open-source package.

AI usage disclosure

No AI tools were used in developing the software included in this release. In ongoing development work, Claude Code has been used for code assistance, and all AI-assisted code is carefully reviewed by the authors.

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